

Adversarial Networks in Attacks

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 19: AI

1. Why Adversarial ML Changes the Threat Model

Traditional security controls are engineered components: a firewall rule either blocks a packet or it does not. Machine learning-based controls — malware classifiers, phishing detectors, anomaly engines — have a different property: they learn a statistical decision boundary, and that boundary can be manipulated. An adversary who understands how a model makes decisions can craft inputs that cross the boundary in the wrong direction. This is the foundational insight of adversarial machine learning, and it forces a shift in how security architects think about AI-based controls. A firewall cannot be fooled into allowing traffic by making the packet look slightly different. A neural network classifier can. The implication is that every ML model deployed as a security control must be threat-modelled as an attack surface in its own right — not merely as a protective layer.

This post-video guide goes deeper than the lecture on three dimensions: the mathematical intuition behind adversarial examples, the operational attack scenarios already observed in production environments, and the layered defensive architecture that the CompTIA SecAI+ exam expects you to recommend.

2. Generative Adversarial Networks: Architecture and Dual Use

Ian Goodfellow introduced the GAN architecture in 2014. The structure is elegant: a Generator network produces synthetic data, and a Discriminator network tries to distinguish synthetic from real. Training proceeds as a minimax game — the Generator improves until the Discriminator can no longer reliably tell real from fake. The practical outcome is a model that produces statistically indistinguishable synthetic examples of whatever the training set contained: faces, fingerprints, network traffic, malware binaries.

For security professionals, the dual-use nature of GANs is fundamental. The same architecture that powers privacy-preserving synthetic dataset generation — a legitimate use — powers deepfake creation, synthetic biometric attacks, adversarial example generation, and GAN-based C2 traffic obfuscation. Evaluating a GAN deployment requires assessing both the intended use and the realistic misuse cases, particularly when the GAN is accessible to external parties or integrated into a system with broad network connectivity.

3. The Mathematics of Adversarial Examples (Intuition)

Understanding adversarial examples does not require graduate-level mathematics, but the SecAI+ exam expects conceptual fluency. An adversarial example is constructed by measuring how the model's output changes in response to small input changes — the gradient. The attacker takes the real input, computes the gradient of the loss function with respect to the input (not the weights), and adds a small perturbation in the direction that maximally increases the model's error. The resulting perturbed input looks identical to the original to a human observer but is classified differently by the model.

The Fast Gradient Sign Method (FGSM), introduced by Goodfellow in 2014, formalises this as: $x_{adv} = x + \epsilon \cdot \text{sign}(\text{gradient})$. Epsilon controls the perturbation magnitude — small enough to be imperceptible, large enough to cross the decision boundary. Projected Gradient Descent (PGD) iterates this step multiple times with a projection back into the allowed perturbation space, producing stronger adversarial examples. Carlini-Wagner (CW) attacks optimise for minimum perturbation rather than fixed epsilon, finding the smallest change needed to fool the model. The practical takeaway: the strength of an adversarial attack depends on attacker knowledge of the model. White-box attacks (full gradient access) are stronger; black-box attacks (output queries only) are weaker but still practically effective due to the transferability phenomenon — adversarial examples often fool models they were not crafted against.

4. Attack Surface Map: Where Adversarial ML Applies

Security Control	ML Component	Adversarial Attack Type	Impact if Successful
Endpoint protection (EDR/AV)	PE malware classifier	Adversarial binary modification	Malware executes undetected
Network IDS/IPS	Anomaly detection model	Adversarial packet crafting	C2 traffic evades detection
Email security gateway	Phishing/spam classifier	Adversarial text perturbation	Phishing email delivered
Facial recognition (access control)	CNN face classifier	Adversarial eyeglasses or patches	Unauthorised physical access
Biometric authentication (fingerprint)	Fingerprint matcher	GAN MasterPrint attack	Authentication bypass
Computer vision surveillance	Person/object detector	Adversarial clothing pattern	Person rendered invisible

5. Adversarial Attacks on Malware Classifiers: Deep Dive

ML-based endpoint protection uses neural networks and gradient-boosted decision trees trained on features extracted from file headers, import tables, byte n-grams, and section entropy. Adversarial attacks against these systems must preserve malicious functionality while shifting the feature vector across the classification boundary — a constraint that makes this harder than adversarial image attacks but not impossible.

- Header manipulation — modifying the MZ header or optional PE header fields to change feature values without affecting execution.
- Section renaming — renaming .text, .data sections to names associated with benign software.
- Byte appending — adding benign-looking byte sequences to unused sections (overlay area) of the PE file.
- Import table padding — adding imports from legitimate DLLs that the malware never actually calls, shifting the import feature vector toward benign.
- Packing/unpacking strategies — applying a benign packer changes the static feature profile entirely; the malicious payload is only visible at runtime.

Published 2023 research demonstrated RL-based adversarial agents that modified PE malware to evade multiple commercial AV engines simultaneously using only the header manipulation and byte appending techniques above — changes that the malware's execution logic completely ignores. The critical defensive implication: static file-based ML classification alone is insufficient; runtime behavioural monitoring that watches actual execution is required as a second layer.

Real-World Incident — Financial Institution Red Team Engagement (2023): A red team used the MalConv-Evade framework (based on academic research) to modify a commodity Remote Access Trojan. Modifications added benign-looking byte sequences to unused PE sections, altering the ML feature vector without changing execution behaviour. The modified binary evaded all seven ML-based endpoint protection products in scope. The engagement revealed that the client's defences relied almost entirely on pre-execution ML classification with inadequate runtime behavioural monitoring — which the RAT triggered only after establishing persistence. Remediation required adding EDR behavioural analytics that operate independently of file-based ML classification.

6. GAN-Powered Synthetic Biometric Attacks

Beyond adversarial examples, GANs enable attackers to generate synthetic data that exploits systematic weaknesses in biometric systems. The MasterPrint attack, demonstrated against commercial fingerprint scanners, leverages a structural property: fingerprint authentication systems compare partial prints against stored templates. A GAN trained to generate synthetic fingerprints that maximise overlap with the statistical distribution of partial prints can produce 'master fingerprints' that match a significant fraction of real users.

In 2024, researchers reported a 76% attack success rate against a commercial fingerprint scanner using GAN-generated synthetic fingerprints. This is not a brute-force attack — it exploits the mathematics of partial fingerprint matching. The defensive countermeasure is liveness detection (presentation attack detection, PAD): biometric systems should verify that the presented sample comes from a live biological source, not a synthesised or replayed artefact. ISO/IEC 30107-3 defines PAD testing requirements for biometric systems.

7. Adversarial Patches and Physical-World Attacks

Adversarial attacks are not confined to digital inputs. Physical adversarial patches — printed patterns designed to fool computer vision models — bridge the ML threat model and physical security. The patch is crafted using standard gradient-based methods, then physically printed and placed in the scene the camera captures.

- Person detection evasion (2022 research) — a specially designed T-shirt pattern caused a state-of-the-art person detection system to fail to detect the wearer in approximately 80% of frames.
- Facial recognition evasion — adversarial eyeglasses frames (printed patterns on standard frames) caused a facial recognition system to misidentify the wearer as a different, authorised individual.
- Object detection evasion — adversarial stickers on stop signs caused autonomous vehicle vision systems to misclassify the sign in controlled research settings.

- License plate spoofing — adversarial patterns overlaid on plates prevented ALPR (Automatic Licence Plate Recognition) systems from reading the plate correctly.

The security implication for physical premises: organisations relying on computer vision for surveillance, access control, or perimeter detection should periodically test their systems with adversarial inputs and not treat the CV model as a trusted control without independent verification layers — human review, additional sensors, or PIN-based secondary authentication.

Core Principle — Adversarial ML Attack Classes: Adversarial attacks target either inference (adversarial examples at prediction time) or training (data poisoning at training time). Both are covered in MITRE ATLAS v2.1. Inference-time attacks require only model access and produce immediate effects. Training-time attacks require access to the data pipeline and produce delayed, potentially invisible effects — embedded backdoors may not trigger until a specific input pattern is presented. The exam distinguishes these by phase: inference-time = Evade ML Model (ATLAS); training-time = Poison Training Data (ATLAS).

8. Data Poisoning and Supply Chain Risk in ML

Data poisoning attacks the training pipeline rather than the deployed model. By inserting carefully crafted malicious examples into the training dataset, an attacker causes the trained model to misbehave on specific inputs while appearing correct on standard test sets. Backdoor poisoning is the most dangerous variant: the poisoned model behaves correctly on all normal inputs but produces attacker-chosen outputs whenever a specific trigger pattern is present.

Published 2024 research demonstrated that poisoning as little as 0.1% of a training dataset — effectively undetectable by standard validation procedures — could embed reliable backdoors in large models. For security applications, this creates a specific supply chain risk: ML models trained on community-sourced threat intelligence data (a common approach in threat detection platforms) inherit the trustworthiness of their contributors. A single malicious contributor can poison shared threat intelligence data to suppress detection of specific malware families.

The 2024 HuggingFace supply chain incident illustrated a related vector: malicious models embedded in the public repository executed code on download via unsafe deserialization in PyTorch pickle files. Organisations using pre-trained models from public repositories must apply checksum validation, provenance verification, and sandboxed fine-tuning — the same supply chain controls applied to open-source code libraries.

9. Defensive Architecture: Layered Countermeasures

No single defence against adversarial ML is comprehensive. The current state of the art requires layering multiple approaches, each addressing different attack vectors:

Defence Category	Mechanism	Strength	Limitation
Adversarial training	Include adversarial examples in training data	Most effective known defence	Computationally expensive; coverage incomplete
Input preprocessing	Smoothing, compression,	Reduces perturbation	Adaptive attackers can

	feature squeezing before inference	effectiveness	bypass
Ensemble methods	Multiple models with different architectures	Reduces transferability of adversarial examples	Cost; may not block targeted attacks
Runtime anomaly detection	Monitor input distributions for statistical outliers	Detects distribution shift	Sophisticated attacks may evade
Certified defences	Mathematical guarantee of robustness within perturbation bound	Strong assurance within bound	Bounds are small; performance cost
Behavioural monitoring (EDR)	Observe execution behaviour, not static features	Catches adversarial binary modifications	Blind to pre-execution ML evasion success

10. Robustness Testing with ART, Foolbox, and CleverHans

The Adversarial Robustness Toolbox (ART), maintained by IBM Research as open source, provides a standardised framework for adversarial attack simulation and robustness evaluation. Key capabilities:

- Attack library — FGSM, PGD, CW, DeepFool, Boundary Attack, and more, covering image, text, audio, and tabular inputs.
- Defence library — adversarial training, feature squeezing, spatial smoothing, JPEG compression defence.
- Robustness metrics — evasion rate under specific attack algorithms and perturbation bounds.
- Evaluation workflow — run attacks against the deployed model, measure what percentage succeed, compare against acceptable risk threshold.

Foolbox and CleverHans are alternative frameworks with slightly different API designs but similar capabilities. The choice between them is typically determined by the ML framework in use (TensorFlow, PyTorch, JAX). For exam purposes: know ART as the IBM-maintained open-source standard; know that adversarial robustness testing should occur before deployment of any ML-based security control, and that results inform the appropriate assurance level for that control in a defence-in-depth stack.

Real-World Incident — GAN MasterPrint Attack on Commercial Fingerprint Scanner (2024): Researchers used a GAN trained on the mathematical structure of partial fingerprint matching to generate 'master fingerprints' — synthetic fingerprints that maximised overlap with the statistical distribution of partial fingerprint templates. Against a commercial fingerprint scanner, the attack achieved a 76% success rate — meaning one of the generated master fingerprints matched over three quarters of test users. The attack exploited the scanner's acceptance of partial matches (a design choice that trades false rejection rate against false acceptance rate) and the structural regularity of fingerprint ridge patterns. Countermeasure: liveness detection per ISO/IEC 30107-3 to reject synthetic or replayed biometric samples.

11. MITRE ATLAS v2.1: The Adversarial ML Taxonomy

MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems) is the adversarial ML counterpart to MITRE ATT&CK. The SecAI+ exam tests knowledge of both and the relationship between them.

- **ATLAS Tactics:** Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Defence Evasion, Discovery, Collection, Exfiltration, Impact — mirroring ATT&CK but applied to ML system components.
- **Evade ML Model (Defence Evasion):** Adversarial examples crafted to cross the model's decision boundary. Applies at inference time.
- **Poison Training Data (Execution/Persistence):** Malicious examples inserted into training data to embed backdoors or suppress detection of specific inputs.
- **Invert ML Model (Discovery):** Model inversion attacks reconstruct training data from model outputs — a privacy violation with security implications.
- **Steal ML Model (Collection):** Repeated queries to extract a functional approximation of the model for offline adversarial attack development.
- **Relationship to ATT&CK:** ATT&CK covers attacks on organisational infrastructure; ATLAS covers attacks on AI/ML components. Real attacks can traverse both — an attacker uses OSINT (ATT&CK) to identify which ML models are deployed, then uses ATLAS techniques to evade them.

12. Responsible Use: Ethics of Adversarial ML Research

Adversarial ML research occupies ethically complex territory because the same techniques that enable authorised security testing also directly enable real attacks. Traditional responsible disclosure norms — find a vulnerability, notify the vendor, wait for a patch, then publish — do not translate cleanly to adversarial ML. There is often no patch for an inherently vulnerable model architecture; the 'fix' is retraining with adversarial examples or architectural changes, neither of which is rapid.

- Operate under explicit written authorisation covering adversarial ML testing scope.
- Document findings for vendor notification with specificity about affected model versions and attack conditions.
- Exercise judgment about public release of working attack implementations — release of effective adversarial attack code directly accelerates attacker capability.
- Apply NIST AI RMF 1.0 GOVERN function guidance and ISO/IEC 42001:2023 responsible disclosure principles.
- When conducting adversarial testing of third-party ML components (e.g., a vendor's AV engine), confirm that contractual terms permit this class of testing — standard pen test authorisation may not explicitly cover adversarial ML.

Key Terms Glossary

Term	Definition
Adversarial example	An input to an ML model deliberately perturbed to cause misclassification, while remaining indistinguishable from a legitimate input to a human observer.
GAN (Generative Adversarial	An ML architecture comprising a Generator (produces synthetic

Network)	data) and a Discriminator (distinguishes real from synthetic), trained adversarially.
FGSM (Fast Gradient Sign Method)	A one-step adversarial attack that adds $\epsilon \cdot \text{sign}(\text{gradient})$ to an input to maximise model prediction error.
PGD (Projected Gradient Descent)	An iterative adversarial attack that repeatedly applies FGSM steps with projection back into the allowed perturbation space; stronger than FGSM.
Data poisoning	Inserting malicious training examples into a dataset to cause the trained model to misbehave on specific inputs or embed backdoors.
Backdoor attack	A data poisoning attack that causes the model to produce attacker-chosen outputs whenever a specific trigger pattern is present in the input.
Adversarial training	A defence that includes adversarial examples in the training dataset so the model learns robustness against them.
MITRE ATLAS	Adversarial Threat Landscape for Artificial-Intelligence Systems — the adversarial ML taxonomy counterpart to MITRE ATT&CK.
MasterPrint	A GAN-generated synthetic fingerprint designed to match multiple real users by exploiting partial fingerprint matching statistics.
ART (Adversarial Robustness Toolbox)	IBM Research open-source framework for adversarial attack simulation and ML robustness evaluation.
White-box attack	An adversarial attack with full knowledge of the model architecture, weights, and gradients — strongest attack assumption.
Black-box attack	An adversarial attack using only model outputs (no internal access) — weaker but practical due to adversarial example transferability.
Transferability	The phenomenon where adversarial examples crafted for one model often fool different models with different architectures.
Liveness detection (PAD)	A biometric system capability that verifies the presented sample comes from a live biological source, not a synthetic or replayed artefact. Governed by ISO/IEC 30107-3.

Quick Revision Checklist

- GANs have dual use: legitimate synthetic data generation and adversarial attacks — deepfakes, biometric attacks, adversarial examples.
- Adversarial examples are crafted using the model's gradient; FGSM is the canonical one-step method; PGD iterates for stronger perturbation.
- White-box attacks use full gradient access; black-box attacks use output queries only; transferability makes black-box attacks practical.
- Adversarial binary modifications (header manipulation, byte appending, section renaming) can evade ML-based AV without changing malware execution.
- Data poisoning targets training time; adversarial examples target inference time — know both ATLAS techniques (Poison Training Data vs. Evade ML Model).
- Backdoor attacks embed trigger-activated misbehaviour; 0.1% poisoning can embed reliable backdoors that survive standard validation.

- GAN MasterPrint attacks achieve high fingerprint authentication bypass rates by exploiting partial match statistics; liveness detection (ISO/IEC 30107-3) is the countermeasure.
- Physical adversarial patches fool computer vision systems in the real world — T-shirt patterns evade person detection; adversarial eyeglasses bypass facial recognition.
- Defences: adversarial training (most effective), input preprocessing, ensemble methods, runtime behavioural monitoring, certified defences — no single defence is sufficient.
- ART (IBM Research) is the standard open-source adversarial robustness testing toolkit; use it to measure evasion rates before deploying ML security controls.
- MITRE ATLAS v2.1 is the adversarial ML taxonomy; know Evade ML Model, Poison Training Data, Invert ML Model, and Steal ML Model techniques.
- Responsible adversarial ML testing requires explicit written authorisation; responsible disclosure for adversarial vulnerabilities is ethically complex — there is often no rapid patch.

10 Exam-Based MCQs — Adversarial Networks in Attacks

Test yourself after reading. These questions are written in exam style — scenario-heavy with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: A red team engagement at a financial institution demonstrated that an adversarially modified RAT binary evaded all seven ML-based endpoint protection products in scope. The modifications added benign-looking byte sequences to unused PE sections. The security architect is reviewing the findings. Which action should the security architect recommend FIRST to address this gap?

- A) Increase the training dataset size for the malware classifier
- B) Add runtime behavioural EDR analytics that operate independently of file-based ML classification
- C) Replace the neural network classifier with a rule-based signature engine
- D) Enable FGSM adversarial training on all endpoint agents

Answer: B **Explanation:** B is correct because the engagement revealed that the attacker succeeded specifically because defences depended on pre-execution ML classification without adequate runtime behavioural monitoring. Adding EDR behavioural analytics as an independent layer addresses this gap directly. A is wrong because increasing training data does not address adversarial robustness — the attack exploited the decision boundary, not a data shortage. C is wrong because replacing the ML classifier with signatures would regress detection capability against novel malware — the correct fix is layering, not replacement. D is wrong because FGSM adversarial training is a research-level technique applied during model retraining, not a configuration option on deployed endpoint agents.

Q2. Scenario: An attacker has modified a commodity malware binary by appending benign-looking byte sequences to unused PE sections and renaming internal sections to names associated with legitimate software. The modified binary successfully evades a commercial neural network-based AV engine. Which MITRE ATLAS technique BEST classifies this attack?

- A) Poison Training Data
- B) Steal ML Model
- C) Evade ML Model

D) Invert ML Model

Answer: C **Explanation:** C is correct because modifying malware to evade an ML-based AV classifier is an inference-time attack that crosses the model's decision boundary — the canonical definition of Evade ML Model in MITRE ATLAS under the Defence Evasion tactic. A is wrong because Poison Training Data targets the training pipeline, not the deployed model at inference time. B is wrong because Steal ML Model involves querying the model repeatedly to extract a functional approximation — there is no model extraction in this scenario. D is wrong because Invert ML Model reconstructs training data from model outputs — a privacy attack, not an evasion attack.

Q3. Scenario: A CISO reviews a research paper demonstrating that GAN-generated synthetic fingerprints achieved a 76% attack success rate against a commercial fingerprint scanner used at corporate building entry points. The attack exploited partial fingerprint matching algorithm properties. What is the MOST appropriate immediate remediation to recommend?

- A) Disable the fingerprint scanner and require PIN-only authentication
- B) Implement liveness detection (presentation attack detection) per ISO/IEC 30107-3
- C) Retrain the fingerprint matcher on GAN-generated synthetic fingerprints
- D) Increase the minimum fingerprint match threshold to 99%

Answer: B **Explanation:** B is correct because the MasterPrint attack exploits the partial fingerprint matching algorithm's statistical properties — a structural vulnerability that cannot be fixed by retraining. Liveness detection (PAD) verifies that the presented sample comes from a live biological source, rejecting GAN-generated synthetic fingerprints regardless of their statistical properties. ISO/IEC 30107-3 is the governing standard. A is wrong because disabling fingerprint authentication entirely is disproportionate and operationally disruptive — the goal is to fix the specific vulnerability, not abandon the authentication modality. C is wrong because retraining on adversarial examples might improve robustness against this specific GAN but cannot address the fundamental partial matching vulnerability; liveness detection addresses the root cause. D is wrong because increasing the match threshold reduces the false acceptance rate somewhat but does not address the root cause — GAN-generated master fingerprints are specifically designed to exceed match thresholds.

Q4. What distinguishes a data poisoning attack from an adversarial example attack?

- A) Data poisoning targets inference time; adversarial examples target training time
- B) Data poisoning targets the training pipeline; adversarial examples target the deployed model at inference time
- C) Data poisoning is only effective against neural networks; adversarial examples affect all ML models
- D) Data poisoning requires physical access; adversarial examples require only network access

Answer: B **Explanation:** B is correct and precisely captures the distinction that MITRE ATLAS encodes as separate techniques: Poison Training Data (training-time attack) versus Evade ML Model (inference-time attack). A is wrong — it inverts the correct relationship. C is wrong — both attack types can affect a range of ML model architectures; the statement about neural networks is false. D is wrong — neither attack type is defined by physical vs. network access; both can be conducted remotely depending on the deployment

context.

Q5. Which property of adversarial examples makes black-box attacks practically viable even without gradient access?

- A) Generalisability
- B) Transferability
- C) Portability
- D) Gradient masking

Answer: B **Explanation:** B is correct. Transferability is the phenomenon where adversarial examples crafted against one model often fool different models with different architectures trained on the same task. This allows black-box attackers to craft examples against a locally trained substitute model and use them against the target model. A is wrong — 'generalisability' is not the standard term for this property in adversarial ML. C is wrong — 'portability' is not the standard term either. D is wrong — gradient masking is a (flawed) defence technique, not an attack property; it involves obscuring the gradient to prevent adversarial example crafting, but it does not explain why black-box attacks work.

Q6. *Scenario:* A security engineering team is preparing to deploy a neural network-based malware classifier as the primary pre-execution detection layer in their EDR platform. Which framework should the team use to measure the malware classifier's adversarial evasion rate before production deployment?

- A) MITRE ATT&CK Navigator
- B) OWASP LLM Top 10
- C) IBM Adversarial Robustness Toolbox (ART)
- D) NIST SP 800-115

Answer: C **Explanation:** C is correct. ART is specifically designed for adversarial attack simulation and ML robustness evaluation — it provides standardised attack implementations and evasion rate metrics for ML models including malware classifiers. A is wrong — ATT&CK Navigator is a visualisation tool for ATT&CK coverage mapping, not an adversarial ML testing framework. B is wrong — OWASP LLM Top 10 covers risks specific to large language model applications, not malware classification models. D is wrong — NIST SP 800-115 is a technical guide for information security testing and assessment (penetration testing), not adversarial ML robustness testing.

Q7. Which NIST AI RMF 1.0 function explicitly addresses adversarial inputs as a risk category?

- A) MANAGE 2.4
- B) GOVERN 1.2 and MAP 1.5
- C) MEASURE 3.1
- D) RESPOND 1.1

Answer: B **Explanation:** B is correct. NIST AI RMF 1.0 addresses adversarial ML explicitly under GOVERN 1.2 (policies and procedures for AI risk governance) and MAP 1.5 (mapping adversarial inputs as a risk

category for AI-reliant systems). A, C, and D refer to other NIST AI RMF sub-categories that do not specifically address adversarial inputs — they may relate to risk management, measurement, and response generally, but GOVERN 1.2 and MAP 1.5 are the specific adversarial ML references that appear in the course material and are exam-relevant.

Q8. What is the PRIMARY risk introduced by sourcing pre-trained models from public repositories such as HuggingFace?

- A) The models may be too large to run on corporate hardware
- B) Backdoors or malicious code in the foundation model propagate to all fine-tuned derivatives
- C) Transfer learning reduces model accuracy below acceptable thresholds
- D) Public models do not support adversarial training

Answer: B **Explanation:** B is correct. The supply chain risk of pre-trained models is that vulnerabilities — backdoors, embedded malicious behaviour, or unsafe serialisation payloads — in a foundation model are inherited by all downstream fine-tuned models. The 2024 HuggingFace incident, where malicious models executed code on download via unsafe PyTorch pickle deserialisation, demonstrated this concretely. A is wrong — hardware compatibility is an operational concern, not a security risk. C is wrong — transfer learning typically improves accuracy on specific tasks; accuracy reduction is not the primary security risk. D is wrong — whether a public model supports adversarial training is an architectural choice, not an inherent risk of public repositories.

Q9. An attacker successfully causes a person detection camera system to fail to detect a person by wearing a specially printed T-shirt pattern. Which category of attack does this represent?

- A) Data poisoning against the surveillance system's training pipeline
- B) A physical adversarial patch attack against a deployed computer vision model
- C) Model inversion to reconstruct training data
- D) A GAN-based synthetic image injection into the camera feed

Answer: B **Explanation:** B is correct. Physical adversarial patches — patterns printed and worn or placed in a scene — are crafted to cause deployed computer vision models to misclassify the scene when captured by a camera. This is an inference-time attack in the physical domain: the adversarial perturbation is in the physical world (the T-shirt pattern) rather than applied digitally. A is wrong — data poisoning requires access to the training pipeline; wearing a T-shirt affects only the deployed model at inference time. C is wrong — model inversion reconstructs training data from model outputs; it does not cause misclassification. D is wrong — a GAN injection attack would require digital access to the camera feed, not a physical artefact worn by the attacker.

Q10. Which combination of controls BEST addresses the adversarial ML risk to an ML-based phishing email classifier?

- A) Increase training dataset size and deploy DMARC
- B) Apply adversarial training, use an ensemble of classifiers, and monitor input distributions for statistical anomalies

- C) Replace the ML classifier with a rule-based signature system
- D) Enable full gradient masking on the deployed classifier

Answer: B **Explanation:** B is correct because it applies three complementary defensive layers: adversarial training (most effective known defence — teaches the model to resist perturbations), ensemble methods (reduces transferability — adversarial examples crafted for one model often don't fool others), and runtime anomaly detection (detects inputs with statistical properties inconsistent with normal email). Together these represent a layered defence-in-depth approach. A is wrong — DMARC addresses email sender authentication, not adversarial manipulation of the classifier; increasing dataset size alone does not improve adversarial robustness. C is wrong — reverting to rule-based signatures would regress detection of novel phishing variants; the correct approach is hardening the ML classifier, not abandoning it. D is wrong — gradient masking is widely considered a weak and brittle defence; adaptive attackers can bypass it, and it does not provide the layered protection needed.